C Proofs for Embedding Scheme

Theorem 6 (Correctness). *Let $m \in \{0,1\}^n$ be the hidden message partitioned into $r = \lceil n/l \rceil$ chunks, E be an embedding function, and H be a locally sensitive hash function that outputs l bits where $H(E(x)) = H(E(x'))$ if distance$(E(x), E(x')) \leq \delta$ for $x, x' \in \{0,1\}^n$ and $\Pr[H(y) = 1] = 1/2$ for a randomly sampled $y \in \{0,1\}^n$. Additionally, assume the language model being used produces enough entropy in each sample that the encoder can always sample a valid token. If for each chunk m_i, the encoder outputs a candidate text x such that*

$$H(E(x)) = m_i,$$

then the decoding algorithm recovers the entire message m with overwhelming probability (i.e., the error probability is negligible in the security parameter λ).

Proof. For a fixed chunk m_i, since the LSH function is balanced, a randomly generated candidate text x from the language model satisfies

$$\Pr\big[H(E(x)) = m_i\big] = \frac{1}{2^l}.$$

Thus, the expected number of samples needed is 2^l. Given the high–entropy assumption of the language model, the probability that the encoder fails to find a valid candidate within a polynomial number of samples is negligible. By applying a union bound over all r chunks, the overall failure probability remains negligible when r is polynomial in λ. Therefore, an honestly encoded message is decoded correctly with overwhelming probability.

Theorem 7 (Undetectability). *Assume that the LSH function H evenly splits the message space (i.e., each l–bit outcome occurs with probability $1/2^l$) and that the PRF used in the perturbation is keyed so that its output is unpredictable. Then, provided that the language model produces high–entropy candidate texts, the rejection sampling procedure yields stegotexts that are statistically indistinguishable from naturally generated texts.*[5]

Proof. The encoder repeatedly samples candidate texts x until $H(E(x)) = m_i$ for the desired chunk m_i. Since H is balanced, $H(E(x)) = m_i$ happens with probability $1/2^l$, and conditioning on this event does not significantly alter the underlying distribution of x. Additionally, the perturbations induced by the keyed PRF (which determines the split) are unpredictable to PPT adversaries. Therefore, the distribution of sampled texts remains statistically close to the natural distribution of texts generated by the language model. Therefore, no efficient adversary can distinguish between stegotexts and naturally generated texts with non–negligible advantage.

Theorem 8 (Robustness). *Let $m = (m_1, \ldots, m_r)$ be a hidden message, and let $x = (x_1, \ldots, x_r)$ be the covertext in which the r chunks of m are respectively embedded. Suppose an adversary obtains x and produces*

$$x' = (x'_1, \ldots, x'_r)$$

by paraphrasing some subset of the covertext chunks, subject to the condition

$$\text{distance}(E(x_j), E(x'_j)) \leq \delta \quad \textit{for each paraphrased chunk } x'_j$$

[5] We ignore low-entropy cases in this proof. In practice, this is handled by sampling incorrect candidates and relying on error correction during recovery.

where $j \in \{1 \dots n\}$. Then, by the design of the LSH function H, the decoding algorithm recovers all r chunks of m with overwhelming probability.

Proof. By strong robustness (Definition 5), whenever a chunk x_j is paraphrased into x'_j with

$$\text{distance}(E(x_j), E(x'_j)) \leq \delta,$$

the LSH function H is designed so that

$$H(E(x'_j)) = H(E(x_j))$$

with probability at least $1 - \nu(\lambda)$ for some negligible function ν. Consequently, the decoded chunk m_j remains correct with probability at least $1 - \nu(\lambda)$ for each $j \in \{1 \dots n\}$.

Define the event F_j to be "chunk m_j fails to decode after paraphrasing." Then

$$\Pr[F_j] \leq \nu(\lambda).$$

Let $F = (F_1 \cup F_2 \cup \cdots \cup F_r)$ be the event that at least one chunk fails. By the union bound,

$$\Pr[F] = \Pr(F_1 \cup \cdots \cup F_r) \leq \sum_{j=1}^{r} \Pr[F_j] \leq r \cdot \nu(\lambda).$$

Hence the probability of successfully decoding all r chunks simultaneously is

$$\Pr(\text{all chunks decode correctly}) \geq 1 - r \cdot \nu(\lambda).$$

Since r is polynomial in the security parameter λ, and $\nu(\lambda)$ is negligible, the product $r \cdot \nu(\lambda)$ is also negligible. Therefore, with overwhelming probability, all chunks of m are recovered correctly from the paraphrased covertext x'. This establishes the scheme's robustness against paraphrasing of any subset of chunks.

D PCA Prompt

We trained a variant of our PCA LSH over a cleaned variant of the Enron corporate email dataset called the Enronsent corpus. [34]. The following prompt is designed to be used with this model and can be found at `https://github.com/NeilAPerry/robust_steganography`.

"""
I wanted to follow up regarding the implementation timeline for the new risk management system. Based on our initial assessment, we'll need to coordinate closely with both IT and Operations to ensure a smooth transition. Please review the attached documentation when you have a moment.

After consulting with the development team, we've identified several key milestones that need to be addressed before proceeding. The current testing phase has revealed some potential integration issues with our legacy systems, particularly in the trade validation module. We're working on implementing the necessary fixes and expect to have an updated timeline by end of week.

Given the complexity of these changes, I believe it would be beneficial to schedule a stakeholder review meeting. We should include representatives from Risk Management, IT Operations, and the Trading desk to ensure all requirements are being met. I've asked Sarah to coordinate calendars for next Tuesday afternoon.
""""